

Support Vector Machines

A training algorithm for optimal margin classifier – Bernhard E. Boser, Vladimir N. Vapnik

HEART LAB 세미나 1주차 SVM논문 리뷰

HEART LAB 학부 인턴과정 유건

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1. Introduction : Why SVM?

Why SVM?

전역 최적해 보장	Local Minimum 존재 X
일반화 성능 이론	Margin 최대화 + Robust, Outlier 처리
효율성과 해석 가능성	Support Vector만으로 결정 경계 형성

2. Maximum Margin Training

이진 분류의 경우 데이터셋

$$(x_1, y_1), (x_2, y_2) \dots (x_p, y_p)$$

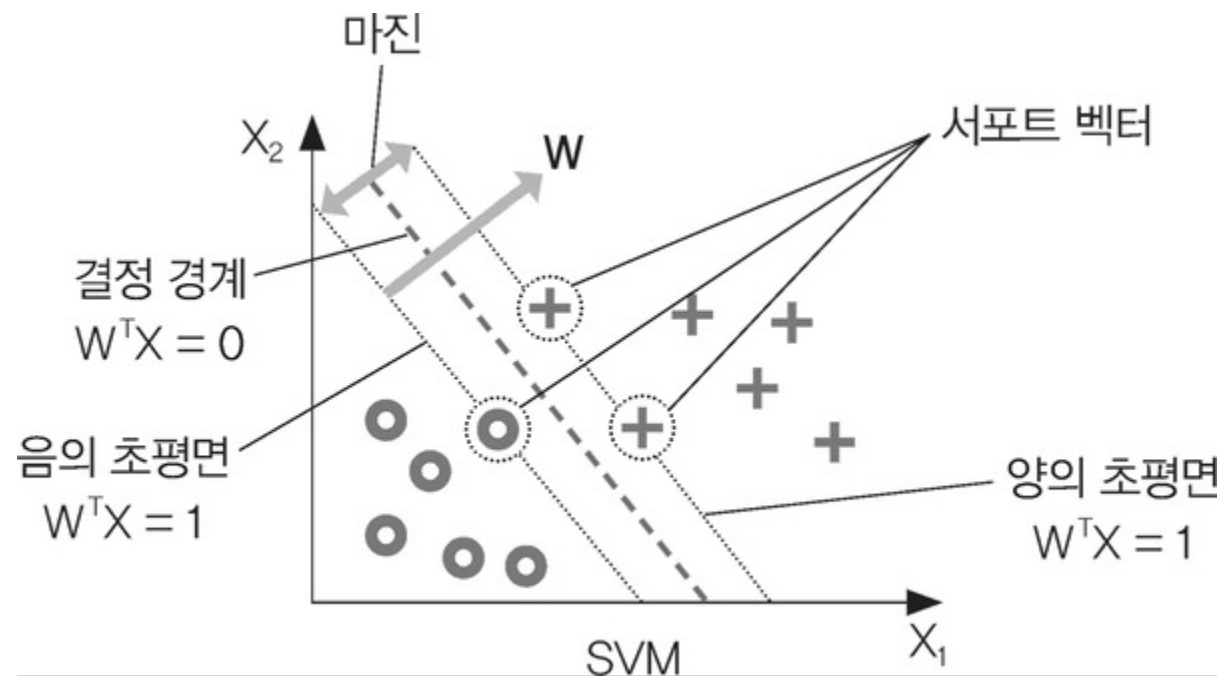
if $x_k \in \text{class A}, y_k = 1$, if $x_k \in \text{class B}, y_k = -1$

1. 직접 공간에서의 결정경계

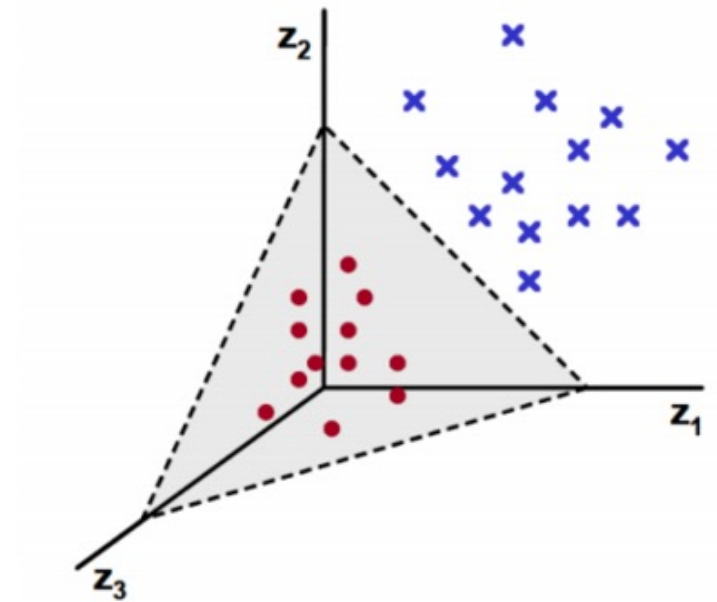
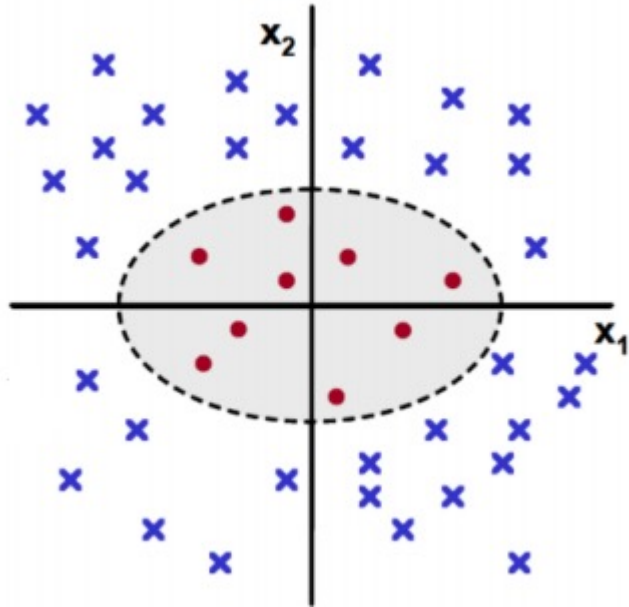
$$D(x) = \sum_{i=1}^N w_i \phi_i(x) + b = W \cdot \phi(x) + b$$

2. 쌍대 공간에서의 결정경계

$$J(\alpha) = \sum_{k=1}^p \alpha_k - \frac{1}{2} \sum_{k=1}^p \sum_{l=1}^p \alpha_k \alpha_l y_k y_l K(x_k, x_l)$$



고차원 매핑 ϕ



비선형 데이터를 고차원 매핑을 통하여 선형 분류를 가능하게 하는 기법

$$\phi(x_1, x_2) = (x_1^2, \sqrt{2}x_1x_2, x_2^2)$$

2.-1 Maximum Margin in Direct Space

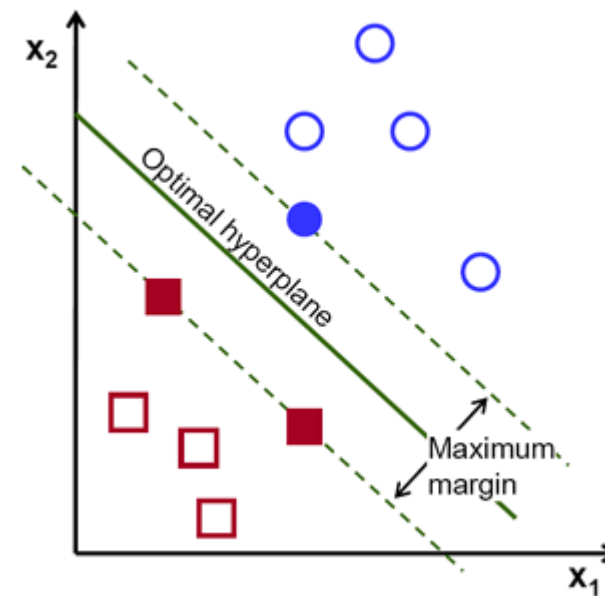
직접공간에서의 최적화

$$D(x) = \sum_{i=1}^N w_i \phi_i(x) + b = W \cdot \phi(x) + b$$

$$\|w\| = M(\text{distance}) = \frac{|w \cdot x_0 + b|}{\|w\|}$$

$$M^* = \max_{W, \|w\|=1} M \text{ subject to } y_k(w \cdot \phi(x_k)) \geq M$$

$$\min_w \frac{1}{2} \|w\|^2 \text{ subject to } y_k(w \cdot \phi(x_k)) \geq 1$$



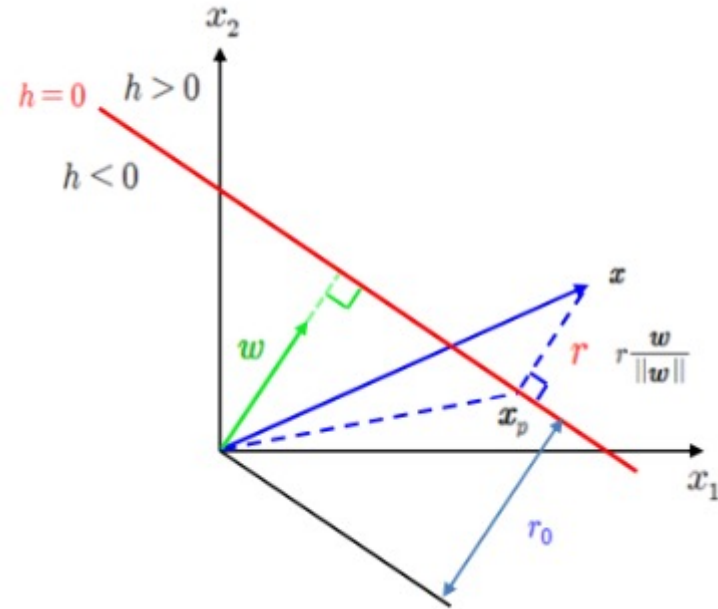
2.-1 Maximum Margin in Direct Space

마진 M 유도과정

$$x_0 - x_p = r \cdot \hat{w}, \quad x_p = x_0 - r \cdot \hat{w}$$

$$w \cdot (x_0 - r \cdot \hat{w}) + b = 0, \quad \hat{w} \cdot ||w|| = w$$

$$r = \frac{|w \cdot x_0 + b|}{||w||}, \quad \text{거리공식이므로 절댓값 붙음}$$



2.-2 Maximum Margin in Dual Space

$$\min_w \frac{1}{2} \|w\|^2 \quad \text{subject to } y_k(w \cdot \phi(x_k)) \geq 1$$



$$\text{maximize } J(\alpha) \quad J(\alpha) = \sum_{k=1}^p \alpha_k - \frac{1}{2} \sum_{k=1}^p \sum_{l=1}^p \alpha_k \alpha_l y_k y_l K(x_k, x_l)$$

2.-2 Maximum Margin in Dual Space

부등식 제약 최적화 풀이: Lagrangian with KKT condition

minimize $f(x)$

subject to $g_i(x) \leq 0 \quad (i = 1, \dots, m)$ & $h_j(x) = 0 \quad (j = 1, \dots, p)$

$$\mathcal{L}(x, \lambda, \nu) = f(x) + \sum_{i=1}^m \lambda_i g_i + \sum_{j=1}^p \nu_j h_j$$

2.-2 Maximum Margin in Direct Space

KKT condition

1. Stationarity (정지조건) $\nabla_x \mathcal{L}(x^*, \lambda^*, \nu^*) = 0$

2. Primal feasibility (원 문제 제약 만족) $g_i(x^*) \leq 0, h_j(x^*) = 0$

3. Dual feasibility (쌍대 변수 조건) $\lambda_i^* \geq 0$

4. Complementary slackness (상보성 조건) $\lambda_i^* \cdot g_i^*(x^*) = 0$

제약이 엄격히 작으면 $\lambda_i^* = 0$

제약이 활성화 되었으면 $\lambda_i^* > 0$

2.-2 Maximum Margin in Dual Space

Dual Space

$$\mathcal{L}(x, \lambda, \nu) = f(x) + \sum_{i=1}^m \lambda_i g_i + \sum_{j=1}^p \nu_j h_j \quad \longrightarrow \quad g(\lambda, \nu) = \inf_x \mathcal{L}(x, \lambda, \nu)$$

minimize $f(x)$

subject to $g_i(x) \leq 0 \quad (i = 1, \dots, m) \ \& \ h_j(x) = 0 \quad (j = 1, \dots, p)$

transfer to maximize $g(\lambda, \nu)$ subject to $\lambda \geq 0$

2.-2 Maximum Margin in Dual Space

Dual Function

$$\min_w \frac{1}{2} \|w\|^2 \quad \text{subject to } y_k(w \cdot \phi(x_k)) \geq 1$$

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2} \|w\|^2 - \sum_{k=1}^p \alpha_k [y_k(w \cdot \phi(x_k) + b) - 1]$$

$$w^* = \sum_{k=1}^p \alpha_k y_k \phi(x_k) \quad \alpha_k \geq 0 \quad \alpha_k [y_k D(x_k) - 1] = 0$$

$$J(\alpha, b) = \sum_k \alpha_k (1 - b \cdot y_k) - \frac{1}{2} \sum_{i=1}^p \sum_{j=1}^p \alpha_i \alpha_j y_i y_j \phi(x_i) \phi(x_j)$$

2.-2 Maximum Margin in Dual Space

$$\frac{\partial \mathcal{L}}{\partial w} = w - \sum_{k=1}^p \alpha_k y_k \phi(x_k) = 0$$

$$\|w\|^2 = \left(\sum_{i=1}^p \alpha_i y_i \phi(x_i) \right) \cdot \left(\sum_{j=1}^p \alpha_j y_j \phi(x_j) \right) = \sum_{i=1}^p \sum_{j=1}^p \alpha_i \alpha_j y_i y_j \phi(x_i) \phi(x_j)$$

2.-2 Maximum Margin in Dual Space

Kernel Function

$$J(\alpha, b) = \sum_k \alpha_k (1 - b \cdot y_k) - \frac{1}{2} \sum_{i=1}^p \sum_{j=1}^p \alpha_i \alpha_j y_i y_j \phi(x_i) \phi(x_j)$$

$$\phi(x_i) \cdot \phi(x_j) = K(x_i, x_j)$$

Linear	$K(x, x') = x^T x'$
Polynomial	$K(x, x') = (x^T x' + c)^d$
RBF(Gaussian)	$K(x, x') = \exp(-\gamma x - x' ^2)$
Sigmoid	$K(x, x') = \tanh(\alpha x^T x' + c)$

2.-2 Maximum Margin in Dual Space

Dual Function + Kernel Trick

$$J(\alpha, b) = \sum_k \alpha_k (1 - b \cdot y_k) - \frac{1}{2} \sum_{k,l} \alpha_k \alpha_l y_k y_l K(x_k, x_l) \quad D(x) = \sum_k \alpha_k y_k K(x_k, x) + b$$

$$b^* = -\frac{1}{2} w^* \cdot (\phi(x_A) + \phi(x_B))$$

$$J(\alpha) = \sum_{k=1}^p \alpha_k - \frac{1}{2} \sum_{k=1}^p \sum_{l=1}^p \alpha_k \alpha_l y_k y_l K(x_k, x_l)$$

$$1. \frac{\partial \mathcal{L}}{\partial \alpha} = 0 \quad 2. \frac{\partial \mathcal{L}}{\partial b} = -\sum_{k=1}^p \alpha_k y_k = 0 \quad 3. \alpha_k \geq 0 \quad 4. \alpha_k [y_k D(x_k) - 1] = 0$$

2.-2 Maximum Margin in Dual Space

수식 b에 대한 증명

$$D(x) = w^* \cdot \phi(x) + b \quad y_A = +1 \text{ 인 } x_A, y_B = -1 \text{ 인 } x_B \text{ 대입}$$

$$w^* \cdot \phi(x_A) + b = 1, w^* \cdot \phi(x_B) + b = -1$$

두 식을 합하면?



$$[w^* \cdot \phi(x_A) + w^* \cdot \phi(x_B)] + 2b^* = 0$$

B에 대해 정리하면?



$$b^* = -\frac{1}{2} w^* \cdot (\phi(x_A) + \phi(x_B))$$

3. Properties of the Algorithm

Unique Solution

Quadratic Objective function	$\min_w \frac{1}{2} \ w\ ^2$
Linear Consistant	subject to $y_k(w \cdot \phi(x_k)) \geq 1$



Quadratic Programming = Quadratic Objective function + Linear Consistant



$H_{ij} = y_i y_j K(x_i, x_j)$ 임의의 벡터 v 에 대해 $v^T H v \geq 0$



Convex Optimization

3. Properties of the Algorithm

Generalization performance

$$1. \text{ Empirical Risk} = \frac{1}{n} \sum_{i=1}^n 1(f(x_i) \neq y_i) \quad \text{True Risk} \leq \text{Empirical Risk} + \text{Complexity Term}$$

$$2. R(f) \leq \frac{m}{n} \quad R(f) = \text{testset에서 Errorrate}$$

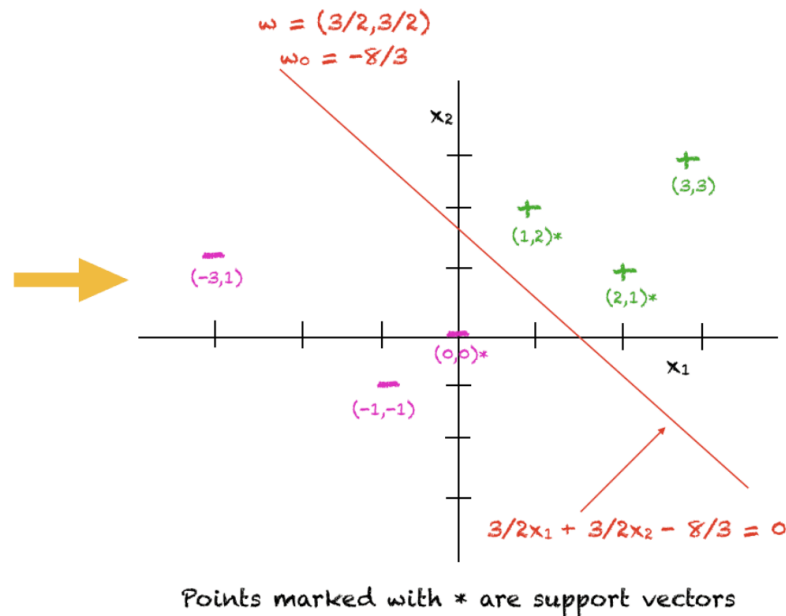
m = 선형 독립인 support vector n = 전체 샘플 수

3. Properties of the Algorithm

Computational Efficiency

$$\max_{\alpha} \sum_{k=1}^p \alpha_k - \frac{1}{2} \sum_{k=1}^p \sum_{l=1}^p \alpha_k \alpha_l y_k y_l K(x_k, x_l) \quad \text{subject to} \quad \sum_{k=1}^p \alpha_k y_k = 0, \alpha_k \geq 0$$

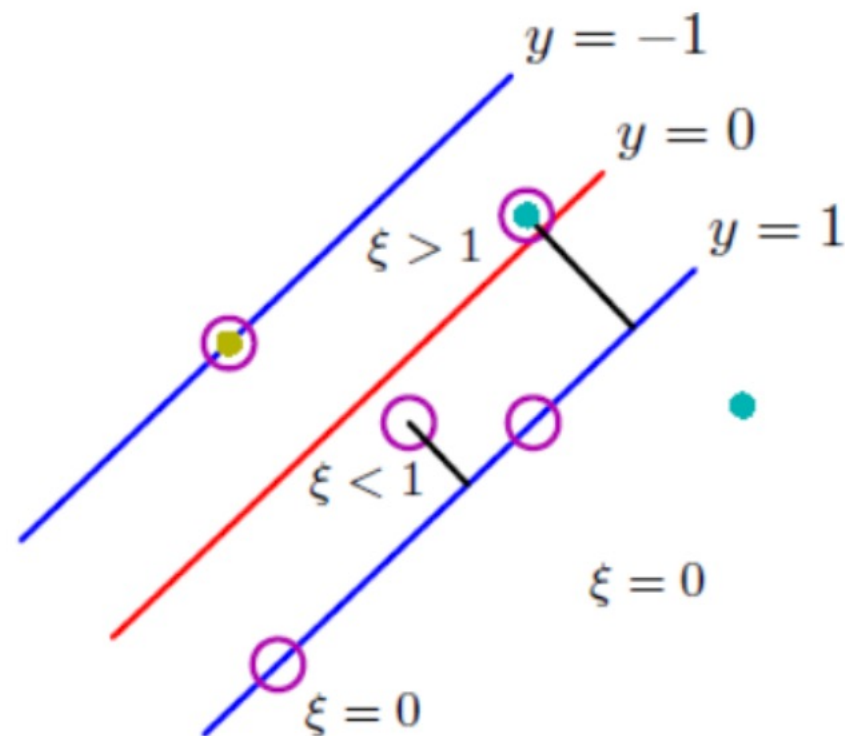
i	Data point x	label t	α
0	(1,2)	+1	0,5
1	(2,1)	+1	0,5
2	(3,3)	+1	0
3	(0,0)	-1	1
4	(-1,-1)	-1	0
5	(-3,1)	-1	0



4. Soft Margin

Slack Variable ξ_i

$\xi_i = 0$	알맞은 라벨이 마진 밖에 위치
$0 < \xi_i < 1$	알맞은 라벨이 마진 내부에 위치
$\xi_i > 1$	잘못 분류된 라벨

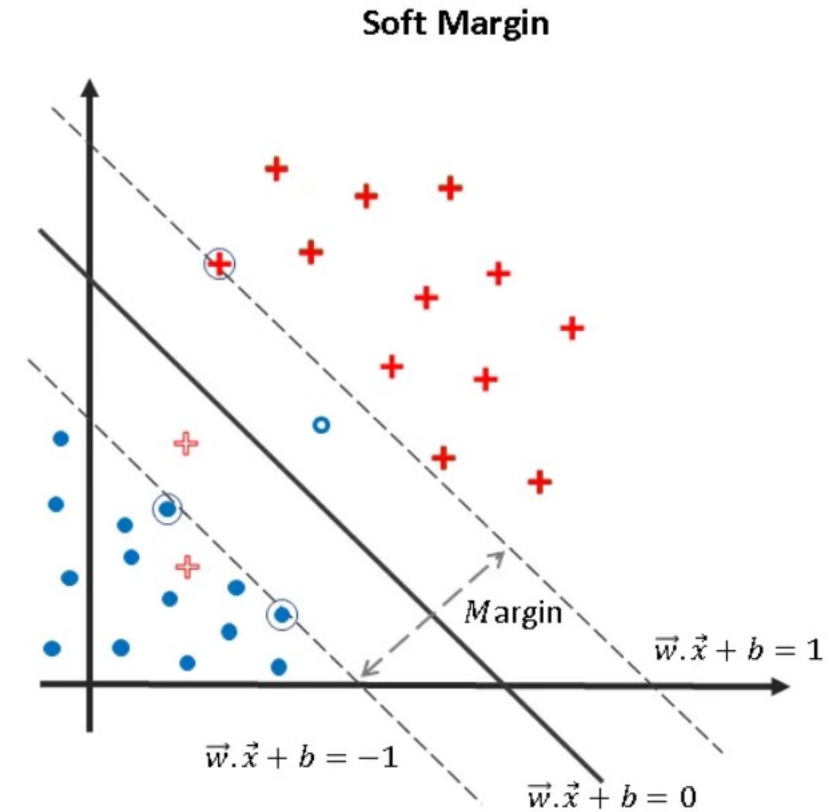


4. Soft Margin

Slack Variable ξ_i

subject to $y_k(w \cdot \phi(x_i) + b) \geq 1 - \xi_i$, $\xi_i \geq 0$

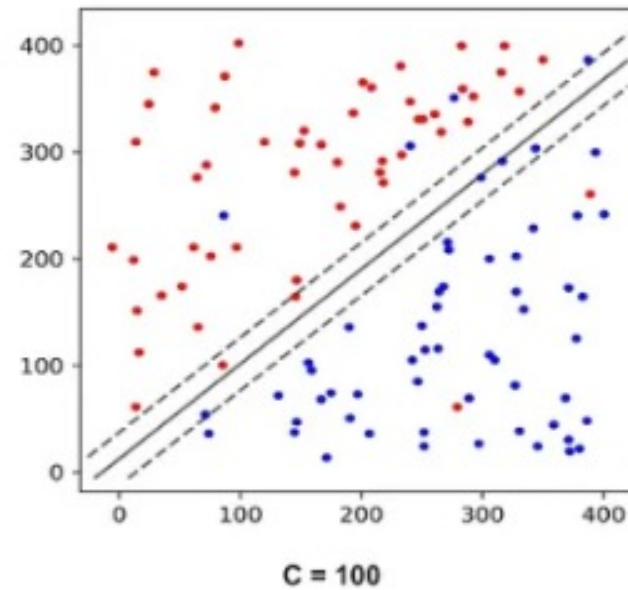
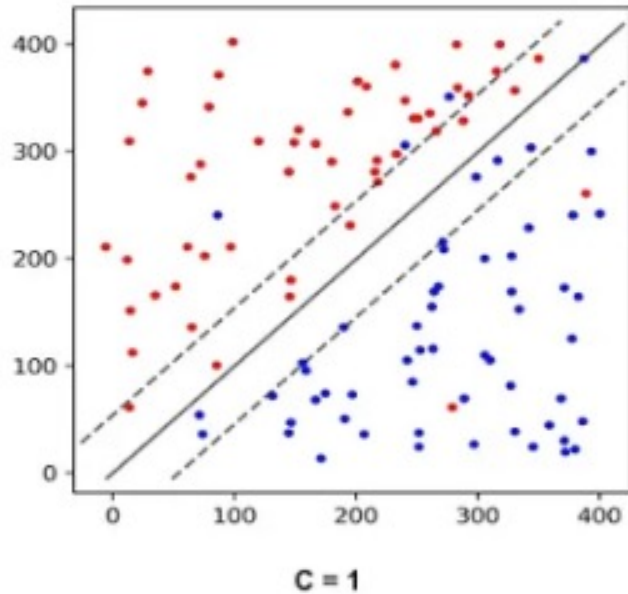
$$\min_{w,b,\xi} \frac{1}{2} ||w||^2 + C \sum_{i=1}^n \xi_i$$



4. Soft Margin

Slack Variable ξ_i 의 계수 C의 최적화 방법

SVM Parameter C



C값이 클수록 Over fitting, C값이 작을수록 Under fitting

4. Soft Margin

Hyperparameter Tuner

GridsearchCV

RandomsearchCV

**Bayesian
Optimization**

4. Soft Margin

Kernel 함수에 따른 Tuner

$K(x, x') = x^T x'$	GridsearchCV, Optuna
$K(x, x') = (x^T x' + c)^d$	Optima, skopt
$K(x, x') = \exp(-\gamma x - x' ^2)$	randomsearchCV, skopt
$K(x, x') = \tanh(\alpha x^T x' + c)$	custom Kernel

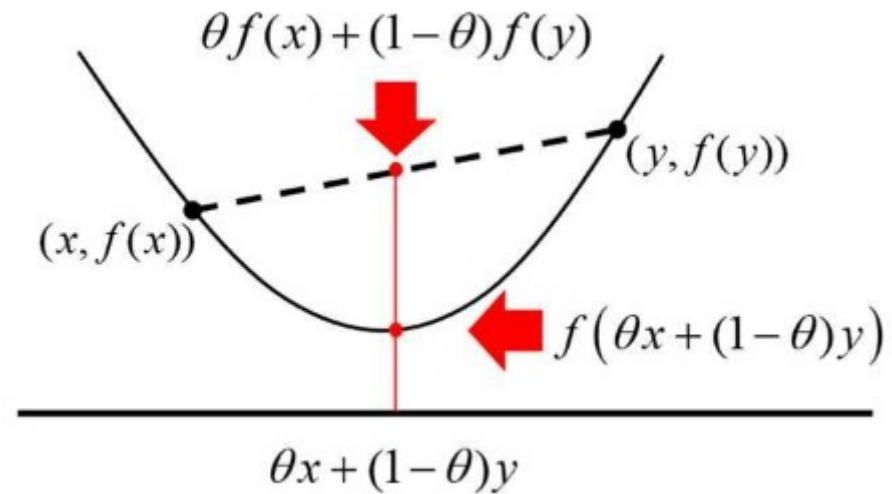
5. Convex Condition

What Does Convex Mean?

$\forall x_1, x_2 \in \text{dom } f, \forall \lambda \in [0, 1] :$

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2)$$

- 함수의 정의역 내 모든 부분이 볼록하다는 의미!
- 접선 불등식 정리에 의해 해당 함수는 Global Minimum을 갖습니다.



5. Convex Condition

Convex condition in Function

1. 목적함수 J 의 Hessian Matrix가 PSD라면 J 또한 PSD여서 convex의 필요충분조건을 충족한다.
2. SVM에서 목적함수는 알파에 대한 일변수함수이므로 헤이시안 행렬은 다음과 같은 1×1 행렬로 나타난다.

$$\nabla^2 f(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

Hessian Matrix

$$\nabla^2 J(\alpha) = \left[\frac{\partial^2 J}{\partial \alpha^2} \right] = H_{ij} \quad (y_i y_j K(x_i, x_j))$$

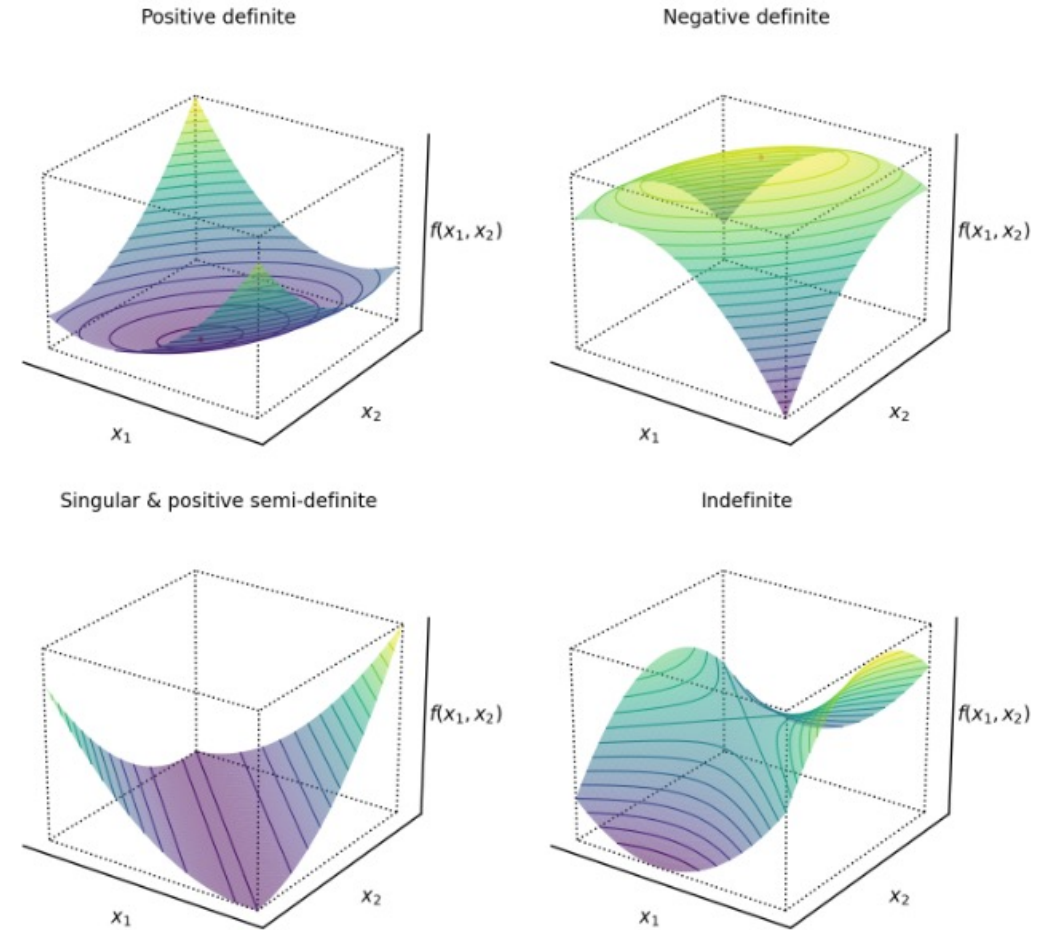
5. Convex Condition

Convex condition in Kernel Matrix

- 기본적으로 머서 정리에 의해 PSD 증명
- 그러나 커널 함수 K의 경우 phi함수 변환해서 증명

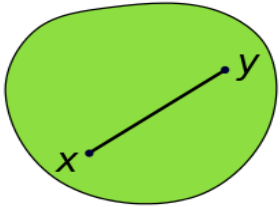
$$\nu^T K \nu = \sum_{i,j} \nu_i \nu_j K(x_i, x_j) = \sum_{i,j} \nu_i \nu_j \phi(x_i)^T \phi(x_j) = \left\| \sum_i \nu_i \phi(x_i) \right\|^2 \geq 0$$

선형 결합의 제곱꼴이므로 항상 0보다 크다

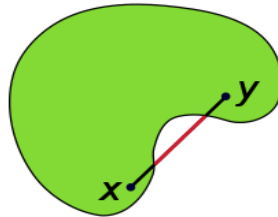


5. Convex Condition

Convex condition in fisible set (제약조건을 만족하는 정의역)



- 제약조건 $x > 0$ 일때
- 조건을 만족하는 $x_1 = 5, x_2 = 10$
- 두 점 사이의 모든 점은 제약조건 만족



- 제약조건 $x_1^2 + x_2^2 = 1$ 일때
- 조건을 만족하는 $m = (-1, 0)$ $n = (0, 1)$ 일때
- 두 점 사이의 모든 점은 제약조건 만족 못함

변수 alpha에 대한 제약조건

$$\alpha_k \geq 0 \quad \alpha_k [y_k D(x_k) - 1] = 0$$

변수 alpha는 선형 결합으로만 이루어져 있으므로 convex set!

5. Convex Condition

Convex condition

커널 함수
Convex

목적함수 J
Convex

제약조건 정의역
Convex

최적화 문제
유일해 보장

Q & A

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감사합니다

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